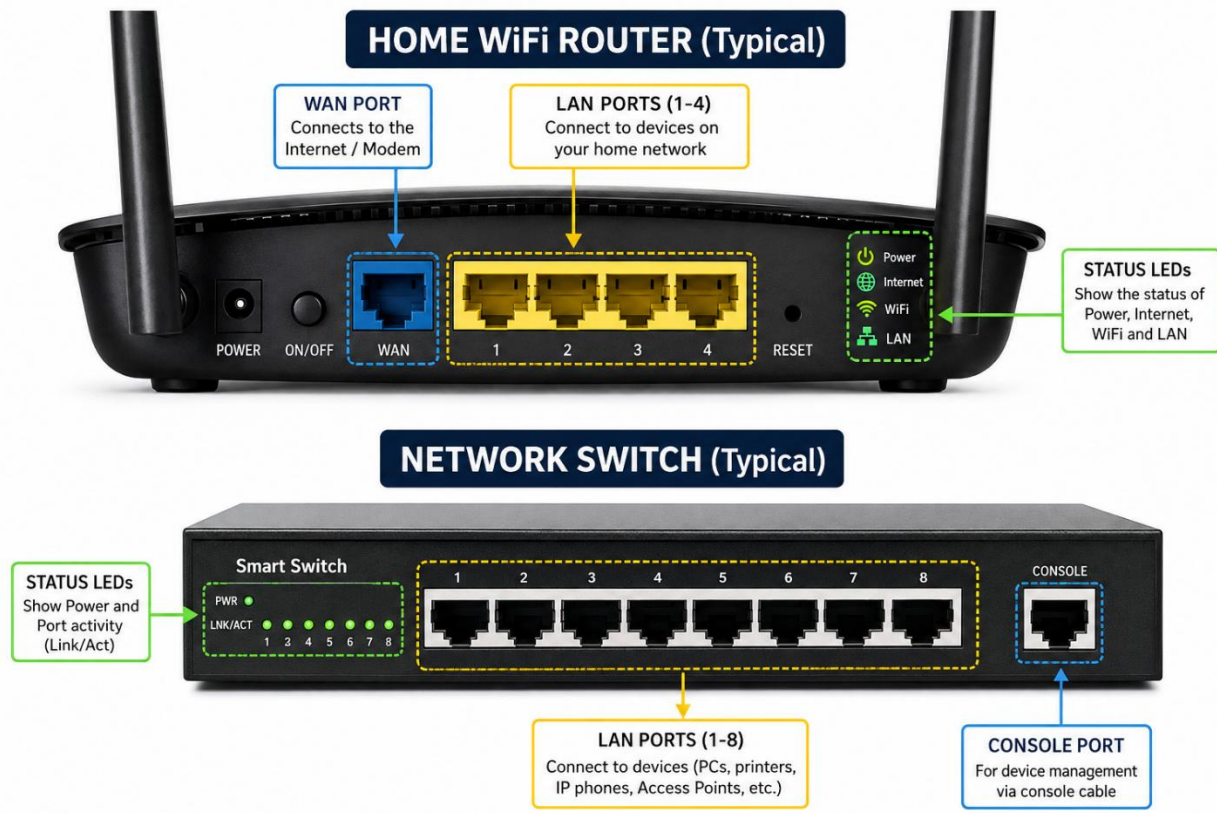


1. Draw a labelled diagram of a typical home WiFi router and a network switch, clearly marking at least 3 different ports (such as WAN, LAN, Console) and the location of status LEDs.

Ans.



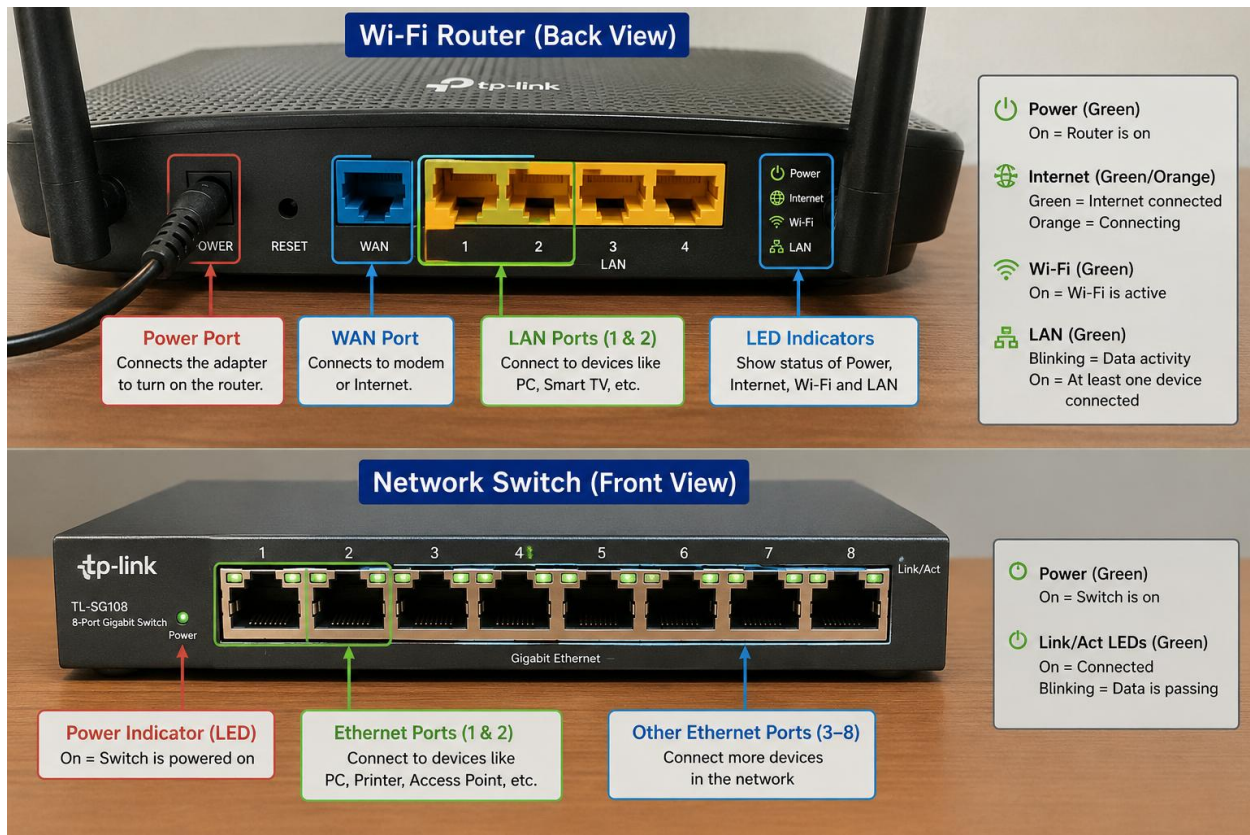
2. List 3 key differences between routers and switches in terms of their main functions and how they handle network traffic. Think about how each device deals with IP addresses, MAC addresses, and the types of networks they connect.

Ans.

Difference	Router	Switch
Main function	Connects different networks and helps devices access the internet.	Connects devices within the same network, such as computers and printers.
Addresses used	Mainly uses IP addresses to decide where network traffic should go.	Mainly uses MAC addresses to send data to the correct device.
Network traffic	Moves data between different networks, such as your home network and the internet.	Moves data between devices on the same local network (LAN).

3. Take a real or virtual image of a router and a switch (from your home, lab, or Google Images), and identify the following on each: Power port, at least 2 Ethernet ports, and any LED indicators. Annotate the image or describe their position and purpose.

Ans.



Router:

- **Power port:** On the back side; connects the power adapter and turns on the router.
- **Ethernet/LAN ports:** Ports 1 and 2 are shown; used to connect devices such as a PC or Smart TV.
- **LED indicators:** Show the status of power, internet, Wi-Fi, and LAN activity.

Network Switch:

- **Power:** The power connection is used to turn on the switch.
- **Ethernet ports:** Ports 1 and 2 are highlighted; they connect computers, printers, access points, and other network devices.

- **LED indicators:** Show whether the switch is powered on and whether a port is connected or sending data.

4. Imagine you are setting up a small gaming café network for IPL streaming and multiplayer matches. Write a short scenario explaining which device (router or switch) you would connect first to the ISP line, and how you would connect 8 gaming PCs to the network. Justify your choices.

Ans. Gaming Café Network Setup

For a small gaming café, I would connect the router to the ISP internet line first. The router brings the internet into the café and manages the network.

Then, I would connect the router to a network switch using an Ethernet cable. After that, I would connect the 8 gaming PCs to the switch, with one Ethernet cable for each PC.

Connection:

ISP → Router → Switch → 8 Gaming PCs

I would use this setup because the router provides internet, while the switch allows many PCs to connect to the same network. Wired connections are also better for gaming because they are usually more stable and have less delay.